

Identification Methods (Review Notes)

1 Unifying Setup: Potential Outcomes and the Identification Problem

Ö Let i index units (firms/individuals/regions) and t index time. A treatment indicator is $D_{it} \in \{0, 1\}$. Potential outcomes are $Y_{it}(1)$ and $Y_{it}(0)$, and the realized outcome is

$$Y_{it} = Y_{it}(0) + D_{it}(Y_{it}(1) - Y_{it}(0)).$$

Ö Estimands (keep interpretation explicit)

- Average Treatment Effect (ATE): $\mathbb{E}[Y(1) - Y(0)]$.
- Average Treatment Effect on the Treated (ATT): $\mathbb{E}[Y(1) - Y(0) \mid D = 1]$.
- Local Average Treatment Effect (LATE): treatment effect for *compliers* (IV/RDD-fuzzy/RKD-type logic).

Ö Why identification is needed A regression like $Y = \beta_0 + \beta_1 X + u$ is causal only if the identifying restriction holds (e.g., conditional mean independence):

$$\mathbb{E}[u \mid X] = \mathbb{E}[u] \quad \Leftrightarrow \quad \text{Cov}(X, u) = 0.$$

Violations (omitted variables, simultaneity, selection, measurement error) create endogeneity.

Ö A consistent reporting style (used throughout these notes) Each design is summarized with the same checklist:

1. **Setup + estimand:** what parameter is being targeted (ATE/ATT/LATE; local vs global).
2. **Key assumptions:** what must be true for causal interpretation.
3. **Baseline estimator:** canonical regression (or moment condition).
4. **Diagnostics + falsification:** what to show in figures/tables.
5. **Inference:** standard errors (clustering level; weak IV; small # clusters).
6. **Interpretation:** internal vs external validity; what the estimate does *not* claim.

2 Difference-in-Differences (DiD)

2.1 Canonical DiD (two groups, two periods)

Setup + estimand Treatment group ($D_i = 1$) vs control ($D_i = 0$), pre vs post ($Post_t \in \{0, 1\}$). The DiD estimand is ATT under parallel trends.

Baseline regression

$$Y_{it} = \alpha + \beta_1 Post_t + \beta_2 D_i + \beta_3 (D_i \times Post_t) + u_{it},$$

where β_3 is the DiD estimate.

Key assumptions

- **Parallel trends:** absent treatment, treated and control would have evolved similarly:

$$\mathbb{E}[Y_{it}(0) - Y_{i,t-1}(0) \mid D_i = 1] = \mathbb{E}[Y_{it}(0) - Y_{i,t-1}(0) \mid D_i = 0].$$

- **No anticipation:** treatment has no effect before implementation.
- **Stable composition:** the relevant sample definition does not change differentially around treatment.

Diagnostics (minimum)

- Plot group means over time to visually assess pre-trends.
- Event-study (dynamic DiD): include leads/lags of treatment to display pre-trends and dynamic effects.
- Placebos: “fake” treatment timing or outcomes that should not respond.

Inference Serial correlation is often severe; clustering is typically required. In panels, cluster at the level of treatment variation (often unit or higher aggregation).

2.2 DiD with multiple periods / staggered adoption

Baseline TWFE specification (common starting point)

$$Y_{it} = \alpha_i + \delta_t + \beta D_{it} + u_{it}.$$

Key issue: heterogeneous / dynamic effects With staggered treatment timing, TWFE can produce non-convex weights and can mix comparisons (treated vs already-treated) that make $\hat{\beta}$ hard to interpret when treatment effects vary by cohort or over time.

Practical fixes (conceptual)

- **Stacked (cohort-specific) DiD:** build event windows around each cohort, use not-yet-treated as controls, and include cohort-by-time and cohort-by-unit fixed effects.

- **Cohort estimands:** estimate $ATT(g, t)$ (group g at relative time t) and then aggregate with transparent weights.
- **Weight diagnostics:** report whether any negative/odd weights arise in the chosen estimator.

''Diagnostics (strongly recommended)

- Event-study plots for each cohort (or aggregated) with clear pre-trend interpretation.
- Sensitivity to window choices (e.g., ± 5 years).
- Robustness to alternative estimators designed for staggered adoption.

3 Regression Discontinuity Design (RDD)

3.1 Core idea

''Setup + estimand Treatment assignment is determined by a running variable x crossing a threshold x_0 . RDD identifies a **local** causal effect at x_0 under continuity conditions.

3.2 Sharp RDD

''Assignment rule

$$D_i = \mathbf{1}\{x_i \geq x_0\}.$$

''Key assumption (local continuity) Potential outcomes are continuous in x at x_0 :

$$\lim_{x \rightarrow x_0^-} \mathbb{E}[Y(0) | x] = \lim_{x \rightarrow x_0^+} \mathbb{E}[Y(0) | x], \quad \lim_{x \rightarrow x_0^-} \mathbb{E}[Y(1) | x] = \lim_{x \rightarrow x_0^+} \mathbb{E}[Y(1) | x].$$

Intuition: absent treatment, Y would not jump *exactly* at x_0 .

''Baseline estimation form (flexible trends) One convenient form is:

$$Y_i = \alpha + \tau D_i + f(x_i - x_0) + D_i \cdot g(x_i - x_0) + \varepsilon_i,$$

where $f(\cdot)$ and $g(\cdot)$ are smooth functions (often local linear or low-order polynomials) and τ is the discontinuity estimate.

''Bias-variance tradeoff Bandwidth choice Δ trades off:

- **Bias:** points far from x_0 are less comparable.
- **Noise:** too narrow a window yields imprecision.

''Graphical analysis (non-negotiable)

- Bin x (no bin straddles x_0), plot binned means of Y against bin midpoints.
- Overlay fitted values from the chosen specification.

3.3 Fuzzy RDD (RDD as IV)

When Crossing x_0 changes the *probability* of treatment but does not deterministically assign it:

$$0 < \lim_{x \rightarrow x_0^+} \Pr(D = 1 \mid x) - \lim_{x \rightarrow x_0^-} \Pr(D = 1 \mid x) < 1.$$

IV formulation Use $Z_i = \mathbf{1}\{x_i \geq x_0\}$ as an instrument for D_i locally:

First stage: $D_i = \pi_0 + \pi_1 Z_i + h(x_i - x_0) + \nu_i$, Second stage: $Y_i = \alpha + \tau D_i + h(x_i - x_0) + \varepsilon_i$.

Interpretation: τ is a **LATE for compliers near the cutoff**.

Internal validity checks

- **Manipulation / sorting:** density test for bunching of x at x_0 .
- **Covariate balance:** baseline covariates should not jump at x_0 .
- **Placebo cutoffs:** no discontinuity where no rule changes.

External validity RDD is local by design; extrapolation away from x_0 is generally not justified without additional structure.

4 Regression Kink Design (RKD)

4.1 Core idea

Setup The *slope* of the treatment assignment function changes at a kink point x_0 (discontinuity in the first derivative), while levels remain continuous.

Identification intuition If the treatment schedule $T(x)$ has a slope change at x_0 , and if the outcome schedule $Y(x)$ also exhibits a slope change at x_0 *for no other reason*, then the causal effect can be recovered from a ratio of slope changes.

Estimation sketch Estimate the change in slope of Y at x_0 and the change in slope of the treatment variable at x_0 ; the causal effect is the ratio. Inference often uses bootstrap or delta method.

Assumptions (high-level)

- No other covariates change slope at x_0 (no confounding kink).
- No manipulation around x_0 .
- Robustness to bandwidth and polynomial order, similar to RDD.

5 Discontinuities for Behavioral/Structural Evidence (Bunching/Notching)

5.1 Goal (different from causal DiD/RDD)

Interpretation Bunching around thresholds is typically interpreted as **behavioral response to incentives** and can be used to learn about mechanisms or structural parameters. It is not automatically a “causal effect of treatment on outcome” in the DiD/RDD sense.

Establishing discontinuity

- Plot the distribution of the running/assignment variable around the threshold.
- Formal density/smoothness tests (histogram-based or local regression-based).

Mechanisms Split samples by predicted incentives or institutional features, then re-test discontinuities to learn which theory best matches the observed pattern.

6 Instrumental Variables (IV)

6.1 Core idea: isolate “good” variation

Setup In $Y = \beta X + u$, endogeneity means $\text{Cov}(X, u) \neq 0$. An instrument Z shifts X but is otherwise unrelated to u .

Two required conditions

- **Relevance:** Z predicts X in the first stage (after controls), i.e. coefficient on Z is non-zero.
- **Exclusion:** $\text{Cov}(Z, u) = 0$; Z affects Y only through X . This is not directly testable and must be defended economically.

2SLS (canonical)

$$\text{First stage: } X = \alpha + \gamma Z + \Gamma W + v, \quad \text{Second stage: } Y = \beta \hat{X} + \Gamma W + \varepsilon.$$

Reduced form: regress Y on Z and W . If the reduced form is essentially zero, the IV is unlikely to matter.

Weak instruments Finite-sample bias grows when the first-stage fit is weak. Report first-stage strength measures and avoid “many weak IVs”.

Interpretation With heterogeneous treatment effects, IV identifies LATE for compliers (those whose X moves with Z). External validity requires care.

Common pitfalls

- “Forbidden regression”: manually plugging fitted values without correct SEs or with nonlinear first stages.

- Using lagged variables as instruments without a strong argument against serial correlation in unobservables.
- Group-average instruments can fail exclusion if group-level unobservables drive both X and Y .

7 Panel Data and Fixed Effects (FE)

7.1 FE as identification against time-invariant omitted variables

Model

$$Y_{it} = \alpha + \beta X_{it} + f_i + u_{it}, \quad \text{Cov}(X_{it}, f_i) \neq 0.$$

The within transformation eliminates f_i :

$$(Y_{it} - \bar{Y}_i) = \beta(X_{it} - \bar{X}_i) + (u_{it} - \bar{u}_i).$$

Identification comes from **within-unit** variation.

Key requirement A strict exogeneity-type condition is needed for causal interpretation: within-unit changes in X must be uncorrelated with within-unit shocks in u (after included controls).

Costs / warnings

- No identification for regressors with no within variation.
- Measurement error can be amplified (attenuation bias worsens).
- FE does not fix time-varying omitted variables; it is not a substitute for design.

Inference Even with FE, serial correlation can remain. Clustering is commonly required (often at the unit level or the level of treatment variation).

8 Matching

8.1 Core idea

Setup Match treated and untreated observations with similar covariates X to approximate the counterfactual.

Identification assumptions

- **Unconfoundedness:** $(Y(0), Y(1)) \perp D \mid X$ (assignment is as-good-as-random conditional on observables).
- **Overlap:** $0 < \Pr(D = 1 \mid X) < 1$ (both treated and control exist for relevant X).

Implementation families

- Nearest-neighbor matching on covariates (distance metrics: Euclidean/Mahalanobis).
- Propensity-score matching: compress X into $\Pr(D = 1 \mid X)$ and match on it.

What matching is (and is not) Matching mainly addresses **observable** imbalance and can be a robustness/sanity check. It does not, by itself, create exogenous variation against unobserved confounding.

9 Structural Estimation vs Reduced Form

9.1 Positioning

Reduced form Targets causal effects of X on Y under an identification strategy (DiD/RDD/IV/FE).

Structural estimation Estimates economic primitives by fitting an explicit economic model and matching moments (GMM/SMM/ML). Identification is achieved through model-implied restrictions (often many at once).

Identification discipline (practical)

- Before estimation: clarify which moments identify which parameters and why.
- Diagnostics: large SEs can indicate weak identification; local identification requires full-rank Jacobian of moments w.r.t. parameters (in GMM/SMM).
- Simulation check: generate fake data from the model and verify the estimator recovers true parameters.

10 Inference Essentials: Standard Errors and Clustering

10.1 Robust vs clustered SE

Robust SE Allows heteroskedasticity but still relies on independence across observations.

Clustered SE Allows arbitrary correlation within clusters. In corporate finance panels, within-firm (or within-industry/state) correlation is common, and ignoring it can strongly understate uncertainty.

Practical clustering rule Cluster at the **most aggregate level at which the key regressor varies**, and assume serial correlation is present unless convincingly ruled out. In DiD, clustering is often crucial.

Small number of clusters Asymptotics depend on the number of clusters, not the number of observations. When clusters are few, inference can be fragile.

11 One-Page Writing Checklist (Design-First)

1. State the estimand (ATE/ATT/LATE; local/global) *before* presenting regressions.
2. Name the identifying assumption in plain language and connect it to the institutional setting.
3. Show the variation: event-time plots (DiD), RD graphs (RDD/RKD), first stage and reduced form (IV).

4. Report diagnostics: pre-trends and weights (DiD), manipulation/balance (RDD), first-stage strength (IV), overlap (matching).
 5. Make inference credible: correct clustering level; discuss weak-IV or small-cluster concerns when relevant.
 6. Limit claims: be explicit about external validity (local effects, compliers, threshold neighborhoods).
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End of review notes.